

Show understanding of binary magnitudes and the difference between binary prefixes and decimal prefixes

9618_w25_qp_13_sc_7.b

QUESTION

(b) A video is made of many bitmap images called frames. 30 frames are recorded every second.

Each frame is 4000 pixels wide by 3000 pixels high. The video records using 16-bit colour depth.

Calculate an estimate for the file size for **one** second of the video in gigabytes.

Show your working.

File size gigabytes

MARK SCHEME

7(b)	1 mark for the working 1 mark for the correct answer Working: $(4000 \times 3000 \times 30 \times 16) / (8 \times 1000 \times 1000 \times 1000)$ $//(4000 \times 3000 \times 30 \times 2) / (1000 \times 1000 \times 1000)$ Answer: 0.72 gigabytes	2
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QUESTION

- (d) A bitmap image has a resolution of 1000 pixels wide by 2000 pixels high. The colour depth is 16 bits.

Calculate an estimate of the file size in megabytes.

Show your working.

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File size megabytes

[2]

MARK SCHEME

6(d)	1 mark for the working 1 mark for the correct answer Working: $(1000 * 2000 * 16) / (8 * 1000 * 1000)$ $// (1000 * 2000 * 2 / (1000 * 1000))$ Answer: 4 megabytes	2
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QUESTION

(d) A bitmap file is downloaded. The image has a maximum of 256 colours and measures 512 pixels wide by 2048 pixels high.

(i) Calculate an estimate of the file size of the bitmap in kibibytes.

Show your working.

Working

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.....

File size of bitmap kibibytes

[2]

MARK SCHEME

7(d)(i)	1 mark for the working 1 mark for the correct answer Working: $(512 * 2048 * 8) / (8 * 1024) // (512 * 2048) / 1024$ $// (2^9 * 2^{11} * 2^3) / (2^3 * 2^{10}) // (2^9 * 2^{11}) / 2^{10}$ Answer: 1024 kibibytes // 2^{10} kibibytes	2
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QUESTION

1 (a) A student paints a picture in an art class.

The student takes a photograph of the picture using a digital camera. The digital camera creates an image with a resolution of 2 million pixels and uses a bit depth of 16 bits.

(i) Calculate the file size of the image created by the digital camera in megabytes (MB).

Show your working.

Working space

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.....

Answer MB

[2]

MARK SCHEME

1(a)(i)	1 mark for correct working: e.g. $2000000 * 16 / (8 * 1000 * 1000)$ 1 mark for answer: 4MB	2
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9618_w24_qp_11_sc_1.a

QUESTION

1 (a) State **one** difference between a tebibyte and a gigabyte.

.....

..... [1]

MARK SCHEME

1(a)	1 mark for: - A tebibyte = 1024 gibibytes / 1 048 576 kibibytes / 2^{40} bytes whereas a gigabyte = 1000 megabytes / 1 000 000 kilobytes / 10^9 bytes - Tebi is binary prefix giga is denary prefix	1
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9618_s24_qp_13_sc_1.a

QUESTION

2

1 (a) Complete the following description.

A kibibyte has a prefix. Three kibibytes is the same as bytes.

A megabyte has a prefix. Two terabytes is the same as gigabytes.

[4]

MARK SCHEME

1(a)	1 mark for each correct answer: • binary • 3072 • denary/decimal • 2000 A kibibyte has a binary prefix. Three kibibytes is the same as 3072 bytes. A megabyte has a decimal/denary prefix. Two terabytes is the same as 2000 gigabytes.	4
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9618_s24_qp_13_sc_2.a

QUESTION

3	
2	A photograph is stored as a bitmap image. (a) The photograph has a resolution of 4000 pixels wide by 3000 pixels high. The bit depth is 4 bytes. Calculate an estimate for the file size of the photograph in megabytes. Show your working. Working Answer megabytes [2]

MARK SCHEME

2(a)	1 mark for working: • $4000 * 3000 * 4$ 1 mark for correct answer: • 48MB	2
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QUESTION

7

A computer stores binary data.

(a)

Tick (✓) **one** box only to identify the **largest** file size.

☐

3300 kibibytes

☐

0.3 megabytes

☐

3 mebibytes

☐

3300 kilobytes

[1]

MARK SCHEME

7(a)	1 mark for: 3300 kibibytes	1
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QUESTION

(b)

Calculate the file size of a bitmap image using the following information:

- image resolution of 2048 pixels wide and 1024 pixels high
- bit depth of 16 bits.

Give your answer in kibibytes. Show your working.

Working

.....

.....

.....

.....

Answer in kibibytes

[2]

MARK SCHEME

2(b)	1 mark for answer. 1 mark for working Answer: 4096 kibibytes Working: File size = $(2048 \times 1024 \times 16) / (8 \times 1024)$	2
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9618_w23_qp_12_sc_3.a

QUESTION

3

(a)

State **one** difference between a kibibyte and a megabyte.

.....

..... [1]

MARK SCHEME

3(a)	1 mark for: Either • kibibyte = 1,024 bytes // 2^{10} bytes and megabyte = 1000 kilobytes // 1 000 000 bytes // 10^3 kilobytes // 10^6 bytes Or • kibi is binary prefix and mega is denary prefix	1
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9618_w23_qp_12_sc_6.c

QUESTION

(c)

A bitmap image has a resolution of 2048 pixels wide and 1024 pixels high.

The image has a bit depth of 10 bits per pixel.

Estimate the file size of the bitmap image in mebibytes. Show your working.

Working

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Estimated file size in mebibytes

[2]

MARK SCHEME

6(c)	1 mark for answer, 1 mark for working Answer: 2.5 mebibytes Working: $(2048 \times 1024 \times 10) / (8 \times 1024 \times 1024)$	2
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9618_s23_qp_11_sc_1.b.ii

QUESTION

(ii) One of the photographs has a bit depth of 8 bytes and an image resolution of 1500 pixels wide and 3000 pixels high.

Calculate the file size of the photograph in megabytes. Show your working.

Working

.....

.....

.....

File size MB

[2]

MARK SCHEME

1(b)(ii)	1 mark for working; 1 mark for answer - Working: $(1500 \times 3000 \times 8) / 1000 / 1000$ - Answer: 36 MB	2
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9618_s23_qp_11_sc_3.d.i

QUESTION

(d) The computer stores data in binary form.

(i) State the difference between a kibibyte and a kilobyte.

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..... [1]

MARK SCHEME

3(d)(i)	1 mark from - Kibibyte is 1024 bytes and kilobyte is 1000 bytes - Kibibyte is binary prefix and kilobyte is denary prefix	1
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9618_w22_qp_13_sc_1.b

QUESTION

- (b) The audio message is recorded with a sampling rate of 50 kHz and a sampling resolution of 16 bits.

The recording is 20 minutes in length.

Calculate the file size of the recording.

Give your answer in megabytes **and** show your working.

Working

.....

.....

.....

Answer megabytes

[2]

MARK SCHEME

1(b)	1 mark for answer; 1 mark for working. Working: Size = 50KHz * (20 × 60) * 16 bits = 50 000 * 1200 * 16 bits // 50 000 * 1200 * 2 bytes = 960 000 000 bits = 120 000 000 bytes = 120 000 kilobytes = 120 megabytes Answer = 120 megabytes	2
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QUESTION

- 1 (a) Draw **one** line from each binary value to its equivalent (same) value on the right.

Binary value

8 bits

8000 bits

1000 kilobytes

1024 mebibytes

8192 bits

1 kibibyte

1 gigabyte

1 byte

1 kilobyte

1 gibibyte

1 megabyte

1 mebibyte

[5]

MARK SCHEME

1(a)	1 mark for each correct line	5
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9618_s21_qp_11_sc_1.a.ii

QUESTION

- (ii) The image is scanned with an image resolution of 1024×512 pixels, and a colour depth of 8 bits per pixel.

Calculate an estimate for the file size, giving your answer in mebibytes. Show your working.

Working

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.....

.....

Answer mebibytes

[3]

MARK SCHEME

1(a)(ii)	1 mark per bullet point for working, 1 mark for answer Working: • $1024 \times 512 = 524\,288$ pixels/bytes • $524288 / 1024 / 1024$ Answer: 0.50 mebibytes	3
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Show understanding of different number systems

9618_w25_qp_13_sc_2.a

QUESTION

- 2 (a) Data in a computer system is represented in binary.

Put **one** tick (✓) in each row to identify the minimum number of bits used to store each example of data.

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139							
16 000 000 unique amplitude values							
an IPv4 address							
256 unique colours							
an IPv6 address							
the denary value 65 000							

[3]

MARK SCHEME

2(a)

3 marks for 6 correct ticks
2 mark for 4 or 5 correct ticks
1 mark for 2 or 3 correct ticks

Example of data	Number of bits						
	4	8	16	24	32	64	128
the hexadecimal value F139			✓				
16 000 000 unique amplitude values				✓			
an IPv4 address					✓		
256 unique colours		✓					
an IPv6 address							✓
the denary value 65 000			✓				

3

9618_w25_qp_13_sc_2.b

QUESTION

(b) Convert the denary number –108 into a 12-bit two's complement binary number.

..... [1]

MARK SCHEME

2(b)	1 mark 1111 1001 0100	1
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9618_w25_qp_12_sc_7.c.i

QUESTION

(c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

(i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

..... [1]

MARK SCHEME

7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
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9618_s25_qp_12_sc_2.b.i

QUESTION

(b) (i) Convert the two's complement binary integer 11100010 into denary.

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..... [1]

MARK SCHEME

2(b)(i)	1 mark for: -30	1
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9618_s25_qp_12_sc_2.b.ii

QUESTION

(ii) Write the smallest and the largest two's complement binary integers that can be represented in 8 bits.

Smallest

Largest [2]

MARK SCHEME

2(b)(ii)	1 mark for smallest, 1 mark for largest Smallest: 1000 0000 Largest: 0111 1111	2
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QUESTION

- (b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

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.....

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.....

Denary value [3]

MARK SCHEME

8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3
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QUESTION

- (b) (i) Convert the denary value into a 12-bit two's complement binary integer.

-196

Answer [1]

MARK SCHEME

3(b)(i)	1111 0011 1100	1
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QUESTION

(ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer [1]

MARK SCHEME

3(b)(ii)	865	1
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QUESTION

7 A computer stores data in binary form.

(a) Draw **one** line from each description to its matching denary value.

Description

Denary value

The smallest integer that can be represented in 8-bit two's complement.

The largest integer that can be represented in 8-bit two's complement.

The largest unsigned integer that can be represented in 8 bits.

–127

127

–255

–128

–256

256

128

255

[3]

MARK SCHEME

7(a)	1 mark for each correct line	3
Description		Denary value
The smallest integer that can be represented in 8-bit two's complement.		-127
		127
The largest integer that can be represented in 8-bit two's complement.		-255
		-128
The largest unsigned integer that can be represented in 8 bits.		-256
		256
		128
		255

9618_s23_qp_12_sc_4.a

QUESTION

- 4 Data in a computer is stored in binary form.
- (a) State the number of unique binary values that can be represented in 16 bits.

..... [1]

MARK SCHEME

4(a)	2^{16} // 65536	1
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9618_s23_qp_12_sc_4.b

QUESTION

(b) Give the 8-bit one's complement representation of the denary number -120 .

Show your working.

Working

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.....

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Answer [2]

MARK SCHEME

4(b)	1 mark for working; 1 mark for answer - Working: $+120 = 0111\ 1000$ - Answer: $1000\ 0111$	2
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9618_s23_qp_11_sc_3.d.ii

QUESTION

(ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

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..... [1]

MARK SCHEME

3(d)(ii)	1001 0110 0100	1
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QUESTION

- (iv) Give the smallest **and** largest two's complement binary number that can be represented using 8 bits.

Smallest

Largest [2]

MARK SCHEME

3(d)(iv)	Smallest: 10000000 Largest: 01111111	2
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QUESTION

- (b) State **two** benefits of using Binary Coded Decimal (BCD) to represent values.

Benefit 1

.....

Benefit 2

..... [2]

MARK SCHEME

9(b)	1 mark for each correct benefit (max 2) Examples: • straightforward to convert to / from BCD and denary • ... so it is less complex to encode and decode for programmers • easier for digital equipment use BCD to display output information • can represent monetary values exactly	2
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9618_w22_qp_12_sc_2.a.i

QUESTION

- 2 (a) (i) Convert the two's complement binary integer into denary.

10010110

Answer

[1]

MARK SCHEME

2(a)(i)	–106	1
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9618_w22_qp_12_sc_2.a.iii

QUESTION

- (iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

.....

.....

Answer

[2]

MARK SCHEME

2(a)(iii)	1 mark for each bullet point: • 149 decimal • 0001 0100 1001	2
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9618_w21_qp_12_sc_4.b

QUESTION

- (b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

..... [1]

MARK SCHEME

4(b)	–51	1
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9618_w21_qp_12_sc_4.d

QUESTION

(d) State why the value in the register cannot be interpreted as a Binary Coded Decimal (BCD).

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 [1]

MARK SCHEME

4(d)	1 mark for: The denary value in each group of 4 bits is greater than 9 // the denary value in each nibble is greater than 9	1
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9618_s21_qp_11_sc_1.c.iii

QUESTION

(iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

.....

.....

.....

.....

Answer (in binary)

[3]

MARK SCHEME

1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 10 = 0000 1010 -10 = 1111 0110 0010 0011 + 1111 0110 0001 1001 11 11	3
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Convert an integer value from one number base/representation to another

9618_w25_qp_13_sc_2.b

QUESTION

(b) Convert the denary number -108 into a 12-bit two's complement binary number.

..... [1]

MARK SCHEME

2(b)	1 mark 1111 1001 0100	1
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9618_w25_qp_13_sc_2.d

QUESTION

(d) Convert the following positive binary integer into hexadecimal.

1110001100111011

..... [1]

MARK SCHEME

2(d)	1 mark E33B	1
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QUESTION

(b) The following table shows some words and corresponding denary values.

Word	Denary value
Computing	55
Science	56
Computers	57
are	58
Brilliant!	59
is	60
Fun!	61
Amazing!	62

The following table shows three bytes of data that have been received.

Use the table to find the corresponding words from the binary values received.

Binary value	00111000	00111100	00111110
Word			

Working

.....

.....

.....

[1]

MARK SCHEME

6(b)	1 mark for correct words in the correct order	1								
<table><tr><td>Binary Value</td><td>0011 1000</td><td>0011 1100</td><td>0011 1110</td></tr><tr><td>Word</td><td>Science</td><td>is</td><td>Amazing!</td></tr></table>			Binary Value	0011 1000	0011 1100	0011 1110	Word	Science	is	Amazing!
Binary Value	0011 1000	0011 1100	0011 1110							
Word	Science	is	Amazing!							

QUESTION

(c) ASCII is another character set. The ASCII value for the character 'h' has the denary value 104.

(i) Write the Binary Coded Decimal (BCD) value for the ASCII character 'h'.

..... [1]

MARK SCHEME

7(c)(i)	1 mark for the correct answer 0001 0000 0100	1
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9618_w25_qp_12_sc_7.c.ii

QUESTION

(ii) Write the hexadecimal value for the ASCII character 'h'.

..... [1]

MARK SCHEME

7(c)(ii)	1 mark for the correct answer 68	1
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9618_w25_qp_11_sc_1.a.i

QUESTION

1 (a) (i) Convert the binary number into hexadecimal.

101100111010

..... [1]

MARK SCHEME

1(a)(i)	B3A	1
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9618_w25_qp_11_sc_1.a.ii

QUESTION

(ii) Convert the denary number into Binary Coded Decimal (BCD).

108

..... [1]

MARK SCHEME

1(a)(ii)	0001 0000 1000	1
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QUESTION

(iii) Convert the 12-bit two's complement binary integer into denary.

Show your working.

111110111100

Working

.....

.....

.....

Denary value

[2]

MARK SCHEME

1(a)(iii)	1 mark for the working 1 mark for the correct denary value Working: Method 1: 1111 1011 1100 Flip the bits 0000 0100 0011 Add 1 0000 0000 0001 + 0000 0100 0100 Method 2: -2048 + 1024 + 512 + 256 + 128 + 32 + 16 + 8 + 4 // -128 + 32 + 16 + 8 + 4 Denary value: -68	2
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9618_w25_qp_11_sc_4.b

QUESTION

(b) The ACC currently contains the following positive binary integer:

1	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---

Write the contents of the ACC after the instruction `XOR &12` is carried out.

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[1]

MARK SCHEME

4(b)	<table><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table>	1	1	1	1	0	0	0	1	1
1	1	1	1	0	0	0	1			

9618_w25_qp_11_sc_4.c

QUESTION

(c) The ACC currently contains the following positive binary integer:

1	1	1	0	0	0	1	1
---	---	---	---	---	---	---	---

Write the contents of the ACC after the instruction `AND #63` is carried out.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

4(c)	<table><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td></tr></table>	0	0	1	0	0	0	1	1	1
0	0	1	0	0	0	1	1			

9618_s25_qp_13_sc_1.c.ii

QUESTION

- (ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

MARK SCHEME

1(c)(ii)	1 mark for each correctly completed space	3																
	<table><tr><th>Character</th><th>Denary</th><th>8-bit Binary</th><th>Hexadecimal</th></tr><tr><td>!</td><td>33</td><td>0010 0001</td><td>21</td></tr><tr><td>L</td><td>76</td><td>0100 1100</td><td>4C</td></tr><tr><td>ü</td><td>252</td><td>1111 1100</td><td>FC</td></tr></table>	Character	Denary	8-bit Binary	Hexadecimal	!	33	0010 0001	21	L	76	0100 1100	4C	ü	252	1111 1100	FC	
Character	Denary	8-bit Binary	Hexadecimal															
!	33	0010 0001	21															
L	76	0100 1100	4C															
ü	252	1111 1100	FC															

9618_s25_qp_12_sc_2.a

QUESTION

- 2 (a) Convert the denary integer 558 into 12-bit binary and hexadecimal.

Binary

Hexadecimal

[2]

MARK SCHEME

2(a)	1 mark for: 0010 0010 1110 1 mark for: 22E	2
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9618_s25_qp_12_sc_2.b.i

QUESTION

(b) (i) Convert the two's complement binary integer 11100010 into denary.

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.....
..... [1]

MARK SCHEME

2(b)(i)	1 mark for: -30	1
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9618_s25_qp_12_sc_7.b.ii

QUESTION

(ii) The ACC currently contains the following binary value.

0	0	0	1	1	1	0	1
---	---	---	---	---	---	---	---

Write the result after the instruction XOR #30 is run.

--	--	--	--	--	--	--	--

[1]

MARK SCHEME

7(b)(ii)	1 mark for: 0000 0011	
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9618_s25_qp_11_sc_3.b.ii

QUESTION

(ii) The Unicode character 'Ɔ' has the binary value: 0010 0111 0110 1110

Convert the binary value for the character 'Ɔ' into denary.

..... [1]

MARK SCHEME

3(b)(ii)	1 mark for: 10,094	1
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9618_s25_qp_11_sc_3.b.iii

QUESTION

(iii) The Unicode character 'Σ' has the hexadecimal value 2140

Convert the hexadecimal code for the character 'Σ' into denary.

..... [1]

MARK SCHEME

3(b)(iii)	1 mark for: 8,512	1
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9618_w24_qp_13_sc_8.a

QUESTION

8 (a) Convert the hexadecimal number 1FAB into denary.

Working

.....

.....

Denary value [1]

MARK SCHEME

8(a)	1 mark for: Denary value: 8107	1
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QUESTION

- (b) Explain how to convert the two's complement integer 10011111 into denary. Give the denary value after conversion.

Explanation

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.....

.....

.....

Denary value [3]

MARK SCHEME

8(b)	1 mark for each bullet point for the method (max 2) e.g. • Flip each bit then add 1 ... • ... method of converting the new binary number into denary • Most significant 1 bit is treated as the corresponding negative denary value ... • ... add the other positive corresponding denary values 1 mark for correct conversion Denary value: -97	3
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QUESTION

- (b) (i) Convert the unsigned binary integer into hexadecimal.

110001100111

Answer [1]

MARK SCHEME

1(b)(i)	1 mark for: C67	1
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9618_w24_qp_11_sc_1.b.ii

QUESTION

(ii) Convert the two's complement binary number into denary.

100110010111

Answer

[1]

MARK SCHEME

1(b)(ii)	1 mark for: -1641	1
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9618_w24_qp_11_sc_1.b.iii

QUESTION

(iii) Convert the Binary Coded Decimal (BCD) into denary.

010101110011

Answer

[1]

MARK SCHEME

1(b)(iii)	1 mark for: 573	1
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QUESTION

(c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

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.....

.....

.....

Answer

[2]

MARK SCHEME

1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 $\begin{array}{r} 0100\ 1010 \\ + 1110\ 1001 \\ \hline (1)0011\ 0011 \end{array}$ Direct subtraction: $\begin{array}{r} 0100\ 1010 \\ - 0001\ 0111 \\ \hline 0011\ 0011 \end{array}$ Answer: 0011 0011	2
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QUESTION

(b) Convert the denary number 241 to hexadecimal.

.....

..... [1]

MARK SCHEME

1(b)	1 mark for correct answer: F1	1
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9618_s24_qp_12_sc_7.b

QUESTION

(b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

.....

.....

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.....

.....

Answer

[3]

MARK SCHEME

7(b)	1 mark each: • Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 • Subtraction method - converting 10 to – 10 and adding // direct subtraction ... • ... correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 ----- (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 -0 0 0 0 1 0 1 0 ----- 0 1 0 1 1 0 1 0 </pre>	3
------	---	---

9618_s24_qp_12_sc_7.c

QUESTION

(c) Convert the hexadecimal number C0F into denary.

Show your working.

Working

.....

.....

Answer

[2]

MARK SCHEME

7(c)	1 mark for working: $1100\ 0000\ 1111 \ //\ 2048 + 1024 + 8 + 4 + 2 + 1$ $//\ (12 * 16^2) + 15 \ //\ (12 * 16 * 16) + 15 \ //\ 3072 + 15$ 1 mark for correct answer: 3087	2
------	--	----------

9618_w23_qp_12_sc_3.b.i

QUESTION

(b) (i) Convert the denary value into a 12-bit two's complement binary integer.

-196

Answer [1]

MARK SCHEME

3(b)(i)	1111 0011 1100	1
---------	----------------	----------

9618_w23_qp_12_sc_3.b.ii

QUESTION

(ii) Convert the Binary Coded Decimal (BCD) into denary.

100001100101

Answer [1]

MARK SCHEME

3(b)(ii)	865	1
----------	-----	----------

9618_w23_qp_12_sc_3.b.iii

QUESTION

(iii) Convert the unsigned binary integer into denary.

000111010110

Answer [1]

MARK SCHEME

3(b)(iii)	470	1
-----------	-----	---

9618_s23_qp_12_sc_4.b

QUESTION

(b) Give the 8-bit one's complement representation of the denary number –120.

Show your working.

Working

.....

.....

.....

Answer [2]

MARK SCHEME

4(b)	1 mark for working; 1 mark for answer - Working: +120 = 0111 1000 - Answer: 1000 0111	2
------	--	---

9618_s23_qp_12_sc_4.c

QUESTION

(c) Convert the hexadecimal number A04 into denary.

Show your working.

Working

.....

.....

.....

Answer [2]

MARK SCHEME

4(c)	1 mark for working; 1 mark for answer - Working: $A04 = (10 * 16^2) + 4$ // $A04 = (10 * 256) + 4$ // $A04 = 1010\ 0000\ 0100$ - Answer: 2564	2
------	---	----------

9618_s23_qp_11_sc_3.d.ii

QUESTION

(ii) Convert the denary number 964 into Binary Coded Decimal (BCD).

.....

..... [1]

MARK SCHEME

3(d)(ii)	1001 0110 0100	1
----------	----------------	----------

9618_s23_qp_11_sc_3.d.iii

QUESTION

(iii) Convert the positive binary integer 11110010 into hexadecimal.

.....

..... [1]

MARK SCHEME

3(d)(iii)	F2	1
-----------	----	----------

9618_w22_qp_13_sc_9.a.i

QUESTION

9 (a) (i) Convert the unsigned binary value into hexadecimal.

10010011

Answer [1]

MARK SCHEME

9(a)(i)	93	1
---------	----	---

9618_w22_qp_13_sc_9.a.ii**QUESTION**

(ii) Convert the unsigned binary value into denary.
10010011
Answer [1]

MARK SCHEME

9(a)(ii)	147	1
----------	-----	---

9618_w22_qp_12_sc_2.a.i**QUESTION**

2 (a) (i) Convert the two's complement binary integer into denary.
10010110
Answer [1]

MARK SCHEME

2(a)(i)	-106	1
---------	------	---

9618_w22_qp_12_sc_2.a.ii**QUESTION**

(ii) Convert the unsigned binary integer into hexadecimal.
10010110
Answer [1]

MARK SCHEME

2(a)(ii)	96	1
----------	----	---

9618_w22_qp_12_sc_2.a.iii

QUESTION

- (iii) Convert the unsigned binary integer into Binary Coded Decimal (BCD). Show your working.

10010101

Working

.....

.....

Answer

[2]

MARK SCHEME

2(a)(iii)	1 mark for each bullet point: • 149 decimal • 0001 0100 1001	2
-----------	---	----------

9618_w22_qp_11_sc_1.a.i

QUESTION

- 1 (a) (i) Convert the unsigned binary integer into denary.

00100111

Answer [1]

MARK SCHEME

1(a)(i)	39	1
---------	----	----------

9618_w22_qp_11_sc_1.a.ii

QUESTION

- (ii) Convert the Binary Coded Decimal (BCD) into denary.

00100111

Answer [1]

MARK SCHEME

1(a)(ii)	27	1
----------	----	---

9618_w22_qp_11_sc_1.a.iii

QUESTION

(iii) Convert the 8-bit two's complement binary integer into denary.

11100111

Answer [1]

MARK SCHEME

1(a)(iii)	-25	1
-----------	-----	---

9618_w21_qp_12_sc_4.a

QUESTION

4 A register stores the following binary number:

1	1	0	0	1	1	0	1
---	---	---	---	---	---	---	---

(a) The binary value in the register represents an unsigned binary integer.

Convert the unsigned binary integer into denary.

..... [1]

MARK SCHEME

4(a)	205	1
------	-----	---

9618_w21_qp_12_sc_4.b

QUESTION

(b) The binary value in the register represents a two's complement binary integer.

Convert the two's complement binary integer into denary.

..... [1]

MARK SCHEME

4(b)	–51	1
------	-----	---

9618_w21_qp_12_sc_4.c

QUESTION

(c) The binary value in the register represents a hexadecimal number.

Convert the binary number into hexadecimal.

..... [1]

MARK SCHEME

4(c)	CD	1
------	----	---

9618_w21_qp_11_sc_1.c

QUESTION

(c) Convert the hexadecimal value F0 into denary.

.....
..... [1]

MARK SCHEME

1(c)	240	1
------	-----	---

QUESTION

(b) Some bit manipulation instructions are shown in the table:

Instruction		Explanation
Opcode	Operand	
AND	#n	Bitwise AND operation of the contents of ACC with the operand
AND	<address>	Bitwise AND operation of the contents of ACC with the contents of <address>
XOR	#n	Bitwise XOR operation of the contents of ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of ACC with the contents of <address>
OR	#n	Bitwise OR operation of the contents of ACC with the operand
OR	<address>	Bitwise OR operation of the contents of ACC with the contents of <address>
<address> can be an absolute address or a symbolic address # denotes a denary number, e.g. #123		

The contents of the memory address 300 are shown:

Bit Number	7	6	5	4	3	2	1	0
300	0	1	1	0	0	1	1	0

(i) The contents of memory address 300 represent an unsigned binary integer.

Write the denary value of the unsigned binary integer in memory address 300.

..... [1]

MARK SCHEME

4(b)(i)	102	1
---------	-----	---

QUESTION

(c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

(i) Write the hexadecimal value for the Unicode character '1'.

..... [1]

MARK SCHEME

6(c)(i)	31	1
---------	----	---

9618_s21_qp_11_sc_1.c.i

QUESTION

(c) One of the colours used in the image has the hexadecimal colour code:

#FC238A

FC is the amount of red, 23 is the amount of green and 8A is the amount of blue in the colour.

(i) Convert the hexadecimal code FC into denary.

..... [1]

MARK SCHEME

1(c)(i)	252	1
---------	-----	---

9618_s21_qp_11_sc_1.c.ii

QUESTION

(ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working

.....

.....

.....

.....

.....

Answer (in binary) [3]

MARK SCHEME

1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer 0010 0011 + 0000 1111 ----- 0011 0010 1 111	3
----------	---	---

Perform binary addition and subtraction

9618_w25_qp_12_sc_7.a

QUESTION

7

(a) The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

1 0 0 0 0 1 1 1

+ 0 0 1 1 1 0 0 1

[1]

MARK SCHEME

7(a)	1 mark for the correct answer	1
	1100 0000	

9618_w25_qp_11_sc_1.b.i

QUESTION

(b)

(i)

The following binary addition is performed using 8-bit registers.

Complete the calculation using binary addition.

1 0 1 1 0 0 1 1

+ 0 1 1 1 1 0 0 0

[1]

MARK SCHEME

1(b)(i)	(1) 00101011	1
---------	--------------	---

9618_w25_qp_11_sc_1.b.ii

QUESTION

(ii) Name and describe the error that can occur when binary addition is performed.

.....

.....

.....

..... [2]

MARK SCHEME

1(b)(ii)	1 mark per bullet point, max 2 marks - An overflow error occurs - The answer cannot be represented in the number of bits available // The answer is larger than the maximum positive number that can be stored in the register // The answer is smaller than the most negative number that can be stored in the register	2
----------	--	----------

9618_s25_qp_13_sc_7.b

QUESTION

(b) Complete the following binary addition. Show your working. Include any overflow bit(s).

$$\begin{array}{r}
 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\
 +\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1 \\
 \hline
 \end{array}$$

[1]

MARK SCHEME

7(b)	1 mark for $ \begin{array}{r} 1\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ +_1\ 1\ 1\ 0\ 1\ 1\ 0\ 0\ 0\ 1\ 1 \\ \hline (1)\ 1\ 0\ 1\ 0\ 0\ 1\ 1\ 0 \end{array} $ Answer: (1) 1010 0110	1
------	--	----------

QUESTION

(c) Subtract the binary number 00011110 from the binary number 01100100 using binary subtraction. Show your working.

[2]

MARK SCHEME

7(c)	1 mark for showing binary subtraction (any method) Direct subtraction: $\begin{array}{r} 0\ 1\ 1\ 1\ 0\ 1\ 1\ 1\ 0\ 0 \\ -0\ 0\ 0\ 1\ 1\ 1\ 1\ 1\ 0\ 0 \\ \hline 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0\ 0 \end{array}$ Adding the two's complement: $\begin{array}{r} 0\ 1\ 1\ 0\ 0\ 1\ 0\ 0 \\ +1\ 1\ 1\ 1\ 1\ 0\ 0\ 0\ 1\ 0 \\ \hline (1)0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \end{array}$ 1 mark for answer 0100 0110	2
------	---	---

QUESTION

(c) Subtract the denary number 23 from the two's complement binary number 01001010

Perform this calculation using binary subtraction.

Show your working.

Working

.....

.....

.....

.....

.....

.....

Answer

[2]

MARK SCHEME

1(c)	1 mark for working 1 mark for answer. Working Using the two's complement and adding: 23 in binary = 00010111 so minus 23 is 11101001 <pre> 0100 1010 + 1110 1001 (1)0011 0011 </pre> Direct subtraction: <pre> 0100 1010 - 0001 0111 0011 0011 </pre> Answer: 0011 0011	2
------	--	----------

9618_w24_qp_11_sc_1.d

QUESTION

(d) State **one** reason why binary addition and subtraction can result in overflow.

.....

..... [1]

MARK SCHEME

1(d)	1 mark from: - The answer cannot be represented in the number of bits available - The answer is larger than the maximum positive number that can be stored in the register - The answer is smaller than the most negative number that can be stored in the register	1
------	---	----------

9618_s24_qp_13_sc_1.c

QUESTION

(c) State what is meant by an **overflow in binary addition**.

.....

..... [1]

MARK SCHEME

1(c)	1 mark for a correct answer: The answer is too long to be represented in the same number of bits as the binary numbers being added	1
------	--	----------

9618_s24_qp_12_sc_7.b

QUESTION

(b) Subtract the denary number 10 from the denary number 100 using binary subtraction.

Show your working.

Working

.....

.....

.....

.....

.....

.....

.....

Answer

[3]

MARK SCHEME

7(b)	1 mark each: • Converting 100 to binary 0110 0100 and 10 to binary 0000 1010 • Subtraction method - converting 10 to – 10 and adding // direct subtraction ... • ... correct answer 0101 1010 Method 1: Converting to -10 and adding: Binary for +10 is 0000 1010 Binary for –10 is 1111 0110 Binary for 100 is 0110 0100 100 + (-10): <pre> 0 1 1 0 0 1 0 0 +1 1 1 1 0 1 1 0 (1) 0 1 0 1 1 0 1 0 Carries: 1 1 0 0 1 0 0 0 </pre> Method 2: Direct Subtraction Borrows: <pre> 0 0 0 1 1 0 1 0 0 1 1 0 0 1 0 0 - 0 0 0 0 1 0 1 0 0 1 0 1 1 0 1 0 </pre>	3
------	--	---

9618_s24_qp_11_sc_7

QUESTION

7 Complete the binary addition. Show your working.

$$\begin{array}{r}
 10011110 \\
 01100001 \\
 + 00011001 \\
 \hline
 \end{array}$$

[3]

MARK SCHEME

7	1 mark each: - Working – carried values clearly indicated - Correct answer 0001 1000 - Overflow clearly indicated as overflow Example: <pre> 1 0 0 1 1 1 1 0 0 1 1 0 0 0 0 1 + 0 0 0 1 1 0 0 1 ----- (1) 0 0 0 1 1 0 0 0 1 1 1 1 1 1 1.....(carries) </pre>	3
---	--	---

9618_s23_qp_11_sc_3.d.v

QUESTION

(v) Add the following two binary integers using binary addition. Show your working.

$$\begin{array}{r}
 10110000 \\
 + 00011011 \\
 \hline
 \end{array}$$

[2]

MARK SCHEME

3(d)(v)	1 mark for working 1 mark for answer <pre> 1 0 1 1 0 0 0 0 + 0 0 0 1 1 0 1 1 ----- 1 1 0 0 1 0 1 1 1 1 </pre>	2
---------	--	---

9618_w22_qp_12_sc_2.b

QUESTION

(b) Perform the following binary addition.

$$\begin{array}{r}
 1\ 0\ 0\ 0\ 1\ 1\ 0\ 0 \\
 +\ 0\ 1\ 0\ 0\ 0\ 1\ 1\ 0 \\
 \hline
 \end{array}$$

[1]

2(b)	1101 0010	1
------	-----------	---

QUESTION

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1 \\ -\ 0\ 1\ 1\ 1\ 0\ 1\ 0\ 1 \\ \hline \end{array}$$

MARK SCHEME

2

	1	10	10	1					
0	10	0	0	10	10				
1	0	1	1	0	0	1	1		
0	1	1	1	0	1	0	1	-	
0	0	1	1	1	1	1	0		

$$\begin{array}{rcccccccc}
 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \\
 \text{Two's complement} & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & + \\
 (1) & \hline
 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\
 & & & & & & 1 & 1 &
 \end{array}$$

9618_w21_qp_12_sc_4.e.i

QUESTION

(e) The binary contents of **two** registers are:

Register 1	0	0	1	1	1	1	0	1
Register 2	0	0	1	0	1	1	0	1

(i) Add the contents of **Register 1** and **Register 2**. Show your working.

Answer [2]

MARK SCHEME

4(e)(i)	1 mark for working, 1 mark for answer <pre> 0011 1101 +0010 1101 ----- 0110 1010 111 1 1 </pre>	2
---------	--	----------

9618_w21_qp_12_sc_4.e.ii

QUESTION

(ii) Subtract the contents of **Register 2** from the contents of **Register 1**. Show your working.

Answer [2]

MARK SCHEME

4(e)(ii)	1 mark for working, 1 mark for answer <pre> 0011 1101 +1101 0011 (two's complement) ----- 0001 0000 1111 111 </pre>	2
----------	--	----------

9618_w21_qp_11_sc_1.b.i

QUESTION

(b) (i) Perform the following binary addition. Show your working.

$$\begin{array}{r} 10101010 \\ + 00110111 \\ \hline \end{array}$$

[2]

MARK SCHEME

1(b)(i)	1 mark for answer 1 mark for working e.g. $\begin{array}{r} 1010\ 1010 \\ 0011\ 0111 \\ \hline 1110\ 0001 \\ 1\ 1\ 1\ 1\ 1\ 1 \end{array}$	2
---------	--	----------

9618_w21_qp_11_sc_1.b.ii

QUESTION

(ii) State how an overflow can occur when adding two binary integers.

.....
..... [1]

MARK SCHEME

1(b)(ii)	The result is a larger number than can be stored in the given number of bits. // The result is greater than <u>255</u>	1
----------	---	----------

QUESTION

(ii) The amount of green in binary is 00100011. This has the denary number 15 added to it to create a second colour.

Add the denary number 15 to the binary number 00100011 and give your answer in binary.

Perform the addition in binary. Show your working.

Working
.....
.....
.....
.....

Answer (in binary)

[3]

MARK SCHEME

1(c)(ii)	1 mark per bullet point • Converting 15 to binary 0000 1111 • Method for addition • Final answer 0010 0011 + 0000 1111 ----- 0011 0010 1 111	3
----------	---	---

QUESTION

- (iii) Hexadecimal 23 in two's complement representation is 00100011. The denary number 10 needs to be subtracted from this value.

Subtract the denary number 10 from the two's complement representation 00100011.

Give your answer in binary. Show your working.

Working

.....

.....

.....

.....

Answer (in binary)

[3]

MARK SCHEME

1(c)(iii)	1 mark per bullet point • Converting -10 to two's complement binary 1111 0110 • Adding values • Final answer 0001 1001 10 = 0000 1010 -10 = 1111 0110 0010 0011 + 1111 0110 ----- 0001 1001 11 11	3
-----------	---	----------

Describe practical applications where Binary Coded Decimal (BCD) and Hexadecimal are used

9618_s25_qp_12_sc_2.c

QUESTION

(c) Give **one** application where Binary Coded Decimal (BCD) is used and justify its use.

Application

Justification

.....

[2]

MARK SCHEME

2(c)	1 mark for application and 1 mark for corresponding justification e.g. - an application that performs financial / banking calculations - because financial transactions use only two decimal places and must be accurate, no accumulating / rounding errors and it is difficult to represent decimal values exactly in normal binary. Or - electronic displays - because visual displays only need to show individual digits - because conversion between denary and BCD is straightforward Or - The storage of the date and time in the BIOS of a PC - because conversion between denary and BCD is more straightforward Or - Barcode systems - because conversion between denary and BCD can be accurately completed	2
------	--	---

QUESTION

(c) Identify **one** practical application of BCD **and** justify why BCD is used in this application.

Application

.....

Justification

.....

.....

.....

[2]

MARK SCHEME

3(c)	1 mark for a correct application 1 mark for a corresponding justification • An application that performs financial / banking calculations • ... because it is difficult to represent decimal values exactly in normal binary and financial transactions use only two decimal places and must be accurate, no accumulating errors Or • Electronic displays, e.g. calculators, digital clocks • ... because visual displays only need to show individual digits • ... because conversion between denary and BCD is easier Or • The storage of the date and time in the BIOS of a PC • ... because conversion with denary is easier	2
------	--	----------

Show understanding of and be able to represent character data in its internal binary form, depending on the character set used

9618_w25_qp_13_sc_1.b

QUESTION

(b) The file names of the sound recordings are stored using the ASCII character set.

Explain how text is represented by the ASCII character set.

.....

.....

.....

..... [2]

MARK SCHEME

1(b)	1 mark per bullet point, max 2 marks	2
	- Each character has a unique code - Each character in the text is replaced sequentially / in order by its code	

9618_w25_qp_13_sc_1.c

QUESTION

(c) Give **two** differences between the ASCII and UNICODE character sets.

1

.....

.....

2

.....

..... [2]

MARK SCHEME

1(c)	1 mark per bullet point, max 2 marks • ASCII uses 7 / 8 bits Unicode can use many more / up to 32 bits • Unicode can represent a wider range of characters including different languages	2
------	---	----------

9618_w25_qp_12_sc_7.b

QUESTION

(b) A computer uses the Unicode character set.

State the number of bits used to store **one** character from the Unicode character set.

..... [1]

MARK SCHEME

7(b)	1 mark for the correct answer 8 // 16 // 32 // 64	1
------	---	----------

9618_s25_qp_13_sc_1.c.i

QUESTION

(c) The student types a report on a computer in a history lesson.

(i) The computer uses the Unicode character set.

Give **two** characteristics of the Unicode character set.

1

.....

2

..... [2]

MARK SCHEME

1(c)(i)	1 mark each to max 2 • 8 / 16 / 32 / bits per character • Represents 2^8 / 2^{16} / etc. characters • Represents every language and other characters such as emojis	2
---------	---	----------

QUESTION

- (ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

MARK SCHEME

1(c)(ii)	1 mark for each correctly completed space	3																
	<table><tr><th>Character</th><th>Denary</th><th>8-bit Binary</th><th>Hexadecimal</th></tr><tr><td>!</td><td>33</td><td>0010 0001</td><td>21</td></tr><tr><td>L</td><td>76</td><td>0100 1100</td><td>4C</td></tr><tr><td>ü</td><td>252</td><td>1111 1100</td><td>FC</td></tr></table>	Character	Denary	8-bit Binary	Hexadecimal	!	33	0010 0001	21	L	76	0100 1100	4C	ü	252	1111 1100	FC	
Character	Denary	8-bit Binary	Hexadecimal															
!	33	0010 0001	21															
L	76	0100 1100	4C															
ü	252	1111 1100	FC															

QUESTION

- (b) The text in the files is stored using the Unicode character set.

- (i) Give **two** advantages of using the Unicode character set instead of using the ASCII character set.

- 1
- 2

[2]

MARK SCHEME

3(b)(i)	1 mark each to max 2 - A wider range of characters can be represented - so characters from more languages can be represented - and symbols such as emojis can be used	2
---------	---	----------

9618_w24_qp_13_sc_6.a

QUESTION

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set

Description

.....

[2]

MARK SCHEME

6(a)	1 mark for identification 1 mark for matching description e.g. - ASCII 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet - UNICODE 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	2
------	--	----------

QUESTION

(d) Computers use character sets when representing characters in binary.

- (i) Complete the table by identifying the number of bits each of the character sets allocates to each character.

Character set	Number of bits
ASCII	
extended ASCII	
Unicode	

[1]

MARK SCHEME

1(d)(i)	1 mark for all 3 answers correct:	1								
	<table><tr><th>Character set</th><th>Number of bits</th></tr><tr><td>ASCII</td><td>7</td></tr><tr><td>extended ASCII</td><td>8</td></tr><tr><td>Unicode</td><td>16/32</td></tr></table>	Character set	Number of bits	ASCII	7	extended ASCII	8	Unicode	16/32	
Character set	Number of bits									
ASCII	7									
extended ASCII	8									
Unicode	16/32									

QUESTION

- (ii) Explain how the word 'Clock' is represented by a character set.

.....

.....

.....

..... [2]

MARK SCHEME

1(d)(ii)	1 mark each: - Each character has a unique binary code - The binary code for each character is stored in sequence	2
----------	--	----------

QUESTION

(c) Give **one** similarity and **two** differences between the ASCII and Unicode character sets.

Similarity

.....

Difference 1

.....

Difference 2

.....

[3]

MARK SCHEME

1(c)	1 mark for similarity, 2 marks for differences Similarity (max 1): • both can use 8 bits • both represent each character using a unique code • Unicode will contain all the characters that ASCII contains // ASCII is a subset of Unicode Differences (max 2): • Unicode can go up to 32 bits per character whereas ASCII is 7 or 8 bits • Unicode can represent a wider range of characters than ASCII • different languages are represented using Unicode, ASCII is only for one language	3
------	---	----------

QUESTION

6 A computer uses the ASCII character set.

(a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII

Extended ASCII

[2]

MARK SCHEME

6(a)	1 mark for each correct answer ASCII = $128 // 2^7$ Extended ASCII = $256 // 2^8$	2
------	---	---

9618_s21_qp_12_sc_6.b

QUESTION

(b) Explain how a word such as 'HOUSE' is represented by the ASCII character set.

.....

.....

.....

..... [2]

MARK SCHEME

6(b)	1 mark per bullet point to max 2 - Each character has its own unique code - Each character in the word is replaced by its code - The codes are stored in the order in the word	2
------	---	---

9618_s21_qp_12_sc_6.c.i

QUESTION

(c) Unicode is a different character set.

The Unicode value for the character '1' is denary value 49.

(i) Write the hexadecimal value for the Unicode character '1'.

..... [1]

MARK SCHEME

6(c)(i)	31	1
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QUESTION

(ii) Write the denary value for the Unicode character '5'.

..... [1]

MARK SCHEME

6(c)(ii)	53	1
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